WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-07-13 to 2026-07-19

Curated articles in political communication and computational social science.

SUMMARY

This week's COMAP Journal Bot scan gathered 249 candidates and curated 13 new papers plus 23 previous-week catch-ups. The clearest throughline is the governance of information and institutions: authoritarian knowledge capture, rebel-founded regimes, wartime hashtag activism, contested police surveillance, scientific consensus, and the political safeguards needed when people delegate decisions to AI.

The methods cluster is equally strong. Agentic AI analysts expose a new multiverse of defensible but divergent results; HALC systematizes prompt construction for automated coding; and work on screen-content distributions, personalization tests, ecological inference, and multiplex networks offers practical measurement infrastructure for computational social science.

- AI appears both as a research instrument and as an object of institutional oversight.
- Political communication papers connect authoritarianism, platform rhetoric, legitimacy, surveillance, and mobilization.
- A few broad-science boundary cases on cognitive measurement, transit allocation, and public governance are retained because their social-science methods or policy implications are plausibly useful to COMAP projects.

HIGHLIGHTS

Media capture, academia capture? Rising authoritarianism and public knowledge capture

JOURNAL OF COMMUNICATION · 2026-07-15

Why it matters · This piece extends the media-capture lens to universities and public knowledge, directly connecting authoritarian power with the institutions that shape what becomes publicly knowable.

Open article →

Many AI analysts, one dataset: Navigating the agentic data science multiverse

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-07-14

Why it matters · Autonomous LLM analysts reproduce substantial analytic dispersion, making multiverse reporting and prompt disclosure concrete requirements for AI-assisted empirical research.

Open article →

Whistleblowers can contain the unethical externalities of human-AI delegation

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-07-16

Why it matters · The experiments identify protected third-party reporting as a practical governance safeguard when people delegate ethically risky decisions to AI systems.

Open article →

When trust building looks like surveillance: Public polarization toward Chicago policing

SCIENCE ADVANCES · 2026-07-17

Why it matters · The study shows that a legitimacy initiative can be read as surveillance and widen racial polarization, linking state communication design to trust and coalition formation.

Open article →

Vengeance or Hope? A Role-Based Analysis of Hashtag Activism for the War in Gaza

POLITICAL COMMUNICATION · 2026-07-14

Why it matters · Cross-lingual X trends and a victim-villain-hero framework reveal asymmetric political rhetoric across partisan and regional contexts.

Open article →

Rebel Victory and Authoritarian (Dis)Order

AMERICAN POLITICAL SCIENCE REVIEW · 2026-07-15

Why it matters · New global data connect the organizational unity of founding rebellions to whether post-rebel authoritarian regimes consolidate or rapidly fracture.

Open article →

Combined signals of scientific and social consensus best mobilize action on societal challenges even if government and industry oppose

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-07-13

Why it matters · Across experiments and a field study, scientific and public consensus jointly sustain action even against elite opposition-a useful result for strategic public communication.

Open article →

Introducing HALC: a general pipeline for the systematic and reliable construction of prompts for automated coding with LLMs in the computational social sciences

COMMUNICATION METHODS AND MEASURES · 2026-07-07

Why it matters · HALC treats prompt construction as a transparent measurement workflow, backed by more than three million LLM requests across varied coding settings.

Open article →

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- Nature Communications
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COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
249	122	127
LATE ADDITIONS CANDIDATES	**CURATED DIGEST BELOW**	**NOTE**
0	13 new, 23 catch-up, 0 late additions	The full collected set above is before manual relevance filtering, while the attached PDF reflects the curated digest.
TIMING NOTE		
COMAP Journal Bot generated this digest on 2026-07-18 for the nominal 2026-07-13 through 2026-07-19 window; articles appearing after collection may be recovered as late additions.		

FULL CURATED DIGEST

New This Week

American Political Science Review

Rebel Victory and Authoritarian (Dis)Order

2026-07-15 · DOI 10.1017/S0003055426101737

KILLIAN CLARKE; ANNE MENG; JACK PAINE

Emory University; Georgetown University; University of Virginia

Abstract. Many authoritarian regimes originating in violent rebellion are exceptionally durable. Yet others collapse quickly after their birth. We argue that the structure of the founding rebellion explains these divergent trajectories by shaping power-sharing dynamics in the new regime and the nature of conflict recurrence. Regimes founded by unified rebellions (single armed group) usually become highly durable autocracies because their members can credibly share power within the same armed organization. By contrast, regimes founded by fractured rebellions (multiple armed groups) often fail quickly because armed rivals must attempt to share power across their organizations, prompting defections and recurrent rebellions. Empirically, we support these arguments by analyzing originally compiled data on every regime founded by successful rebellion worldwide from 1900 to 2020. Neither alternative rebellion characteristics, like social revolution or leftist ideology, nor country-level factors can explain why rebel regimes follow divergent trajectories.

Read article →

Journal of Communication

Media capture, academia capture? Rising authoritarianism and public knowledge capture

2026-07-15 · DOI 10.1093/JOC/JQAG008

Ayala Panievsky

Department of Journalism , City St George's University of London, London,

Abstract. Abstract unavailable.

Read article →

Nature

People use fast and flat simulation to reason about new games

2026-07-15 · DOI 10.1038/S41586-026-10722-1

Katherine M. Collins; Cedegao E. Zhang; Lionel Wong; Mauricio Barba da Costa; Graham Todd; Adrian Weller; Samuel J. Cheyette; Thomas L. Griffiths; Joshua B. Tenenbaum

Abstract. Abstract Games have long been a microcosm for studying planning and reasoning in both natural and artificial intelligence, often focusing on expert-level or even super-human play 1-6 . But real life also pushes human intelligence along a different frontier, requiring people to flexibly navigate decision-making problems that they have never thought about before. Here we use novice gameplay to study how people reason about new problem settings. Through a series of large-scale behavioural studies with over 1,000 participants and 121 two-player strategic board games (almost all novel to our participants), we show that people are systematic and adaptively rational in how they play a game for the first time or evaluate

a game (for example, how fair or how fun it is likely to be) before they have played it even once. We explain these capacities via a computational cognitive model that we call the 'Intuitive Gamer': a model based on mechanisms of fast and flat (depth-limited) goal-directed probabilistic simulation. Our work offers insights into how people rapidly evaluate, act and make suggestions when encountering novel problems, and could inform the design of more flexible and human-like artificial intelligence systems that can determine not just how to solve new tasks but also whether a task is worth thinking about at all.

Read article →

Food systems transformation would reshape global agriculture

2026-07-15 · DOI 10.1038/S41586-026-10775-2

Matthew Gibson; Marina Sundiang; Daniel Mason-D'Croz; Thais Diniz Oliveira; Felicitas Beier; Lauren Benavidez; Astrid Bos; Maksym Chepeliev; Jonathan Doelman; Shahnila Dunston; Shinichiro Fujimori; Tomoko Hasegawa; Petr Havlik; Jordan Hristov; Jonas Jagermeyr; Marta Kozicka; Marijke Kuiper; Page Kyle; Thijs de Lange; Benjamin Leon Bodirsky; Hermann Lotze-Campen; Hermen Luchtenbelt; David Meng-Chuen Chen; Abhijeet Mishra; Christoph Muller; Gerald Nelson; Amanda Palazzo; Ignacio Perez Dominguez; Alexander Popp; Ronald Sands; Marco Springmann; Elke Stehfest; Timothy B. Sulser; Kiyoshi Takahashi; Gianmaria Tassinari; Ferike Thom; Philip Thornton; Kazuaki Tsuchiya; Willem-Jan van Zeist; Hans van Meijl; Dominique van der Mensbrugghe; Detlef Van Vuuren; Hannah H. E. van Zanten; Isabelle Weindl; Keith Wiebe; Xin Zhao; Mario Herrero

Abstract. Abstract Food systems are a major contributor to exceeding planetary boundaries 1-3 and poor quality diets are a key mortality risk globally 4 . Projected population and income growth could exacerbate these challenges 5 . In response, there are calls for transformation towards healthy and sustainable food systems 6-8 . However, the scale and distribution of the impacts of this transformation on agriculture are underexplored. Here we show that, by 2050, the transformation of food systems towards healthy diets (adoption of the EAT- Lancet reference diet), improved productivity and halving of food waste results in a fundamental restructuring of global agriculture, aspects of which break with historical trends. Scenario simulations using a multimodel ensemble of ten global economic models show a 6% median decrease in agricultural land (+1% to 26%) compared with 2020 levels. By 2050, agricultural production would be 17% lower than business-as-usual projections (2% to 32%) and, economically, the value of this production is US$1.6 trillion (26%) lower (+8% to 58%). Within this, the value of livestock production would be substantially lower than current 2050 projections (49% to 83%), while vegetable, fruit, nut and legume production value would increase by 23% (33% to +106%). Results are dependent on the assumed policies to achieve the transformation scenario. We highlight a more active role for food policy to consider the benefits of such a transformation (improved population health and reduced environmental pressures) and navigate the political economy of its impacts.

Read article →

Nature Communications

The economically optimal mix and timing of coastal adaptation in Europe to 2150

2026-07-15 · DOI 10.1038/S41467-026-74042-8

Vanessa Volz; Jochen Hinkel; Daniel Lincke; Lars E. Honsel; Robert J. Nicholls; Remi Thieblemont; Goneri Le Cozannet; Paul Sayers

Abstract. Abstract Coastal adaptation to sea level rise involves combining and sequencing adaptation options into adaptation pathways that include adaptation tipping points, which are moments requiring a switch between options, often due to economic limits. While local case studies show costs depend on the economically optimal timing of actions and adaptation tipping points, this remains unexplored on global and continental scales. We present economically optimal adaptation pathways for Europe's coastal floodplains through 2150. For 95% of the coastline requiring adaptation, the optimal timing for initial actions is immediate due to current economic under-adaptation, a condition in which the costs of additional adaptation are outweighed by the reduction in expected flood damages. By 2150, retreat is economically optimal for 22% of the coastline, incurring lower total costs than protection, which is optimal for 9%. Adaptation tipping points are primarily driven by sea level rise and necessitate a switch from accommodation to either protection or retreat, or from retreat to inland protection

combined with retreat in the foreland. Their timing depends on climate change, occurring 32 years earlier under higher climate change scenarios compared to lower ones.

Read article →

Political Communication

Vengeance or Hope? A Role-Based Analysis of Hashtag Activism for the War in Gaza

2026-07-14 · DOI 10.1080/10584609.2026.2702539

Tay Jeong

Sungkyunkwan University

Abstract. Using cross-lingual international X trends data and a unique role-based analytical framework well-suited for the study of granular textual data, this study examines the partisan and spatial distribution of political rhetoric in hashtag activism about the Gaza War in its early phase. Following a triadic classificatory rubric of roles composed of victim-villain-hero, I find that pro-Palestine hashtag activism was overwhelmingly focused on victimhood, but Muslim-majority countries in the Middle East uniquely exhibited comparably strong expressions of glorified sentiments of heroism. In contrast, pro-Israel hashtag activism was driven by villain-centric messages focusing on the topic of denunciation mixed with the rhetoric of victimhood. Contrary to popular characterizations of the discursive conflict as a parallel clash of victimhood or vengeful emotions, the results reveal stark asymmetries in how wartime militancy was rhetorically framed. I propose a power-centric interpretation of the observed rhetorical asymmetries, which I argue stem from diverging practical needs in a heavily unequal bilateral conflict. I also reflect on regional differences within the global Palestine solidarity movement amidst debates about the cultural praxis of subalternity that have intensified since the outbreak of the Gaza War.

Read article →

Proceedings of the National Academy of Sciences

Response duration tracks confidence

2026-07-17 · DOI 10.1073/PNAS.2612367123

Hanbei Zhou; Rui Zhe Goh; Ian Phillips; Chaz Firestone

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Abstract. The timing of behavioral responses is among the most robust signatures of cognitive processing-with easier, more accurate, and more confident judgments typically made more quickly. But which aspects of response timing carry these signals? Whereas the time taken to initiate a response (e.g., the interval between stimulus onset and keypress) is one of the most popular and informative measures in experimental psychology, another aspect of response timing is its duration (e.g., how long the key is held down before being released). Here, across three preregistered experiments spanning perceptual and higher-cognitive tasks, we demonstrate that response duration reliably predicts subjective confidence, over and above response initiation time. When detecting faces (Experiment 1), judging motion coherence (Experiment 2), and answering trivia questions (Experiment 3), subjects held down a response key longer during trials in which they subsequently reported lower confidence in their judgments, as if making these judgments in a tentative fashion. In all cases, response duration accounted for variance in confidence that was not predicted by response initiation time. Our results suggest that response duration itself carries independent information about cognitive processing, advancing a useful and ubiquitously available tool to complement traditional response time measures in the study of cognition.

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Whistleblowers can contain the unethical externalities of human-AI delegation

2026-07-16 · DOI 10.1073/PNAS.2536668123

Zoe A. Purcell; Nils Kobis; Andrew Samuel; Jean-Francois Bonnefon

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Abstract. Prior work using controlled principal-agent experiments suggests two risks from delegating tasks to AI systems: Human principals are more likely to request profit-maximizing misconduct from AI agents than from human agents, and AI agents are more likely to comply. Here we test whether third-party observers can contain the resulting harm. In an incentivized die-reporting paradigm, principals instructed either a human or an AI agent how strongly to prioritize profit over accuracy, creating potential financial harm to a charity. We first confirm, with human principals (N = 600) and three large language models as AI agents, that delegation to AI produces larger negative externalities than delegation to humans. We then study observers who could pay a personal cost to flag a principal's instruction, canceling the principal's gain in favor of the charity, as a laboratory analogue of whistleblowing. In this observer study (N = 300), the probability of flagging increased with how unethical the principal's request was, but did not depend on whether the request was directed to a human or an AI agent. Because principals made more unethical requests under AI delegation, flagging was more frequent under AI delegation. When combined with agent behavior, this increase in flagging fully neutralized the negative externalities of AI delegation in our experimental setting. These findings support institutional protections for whistleblowers as one potential organizational safeguard against the harms of human-AI delegation.

Read article →

Many AI analysts, one dataset: Navigating the agentic data science multiverse

2026-07-14 · DOI 10.1073/PNAS.2606495123

Martin Bertran; Riccardo Fogliato; Zhiwei Steven Wu

Amazon; Software and Societal Systems Department, School of Computer Science, Carnegie Mellon University

Abstract. Empirical conclusions depend not only on data but also on analytic decisions. Many-analyst studies have quantified this dependence: independent teams testing the same hypothesis on the same dataset regularly reach conflicting conclusions. But such studies require costly human coordination. We show that fully autonomous AI analysts built on large language models (LLMs) can, cheaply and at scale, produce the analytic dispersion observed in human many-analyst studies. In our framework, each AI analyst independently executes a complete analysis pipeline on a fixed dataset and hypothesis; a separate AI auditor screens every run for methodological validity. Across three datasets, AI analyst-produced analyses exhibit substantial dispersion in effect sizes, P -values, and conclusions. This dispersion can be traced to identifiable analytic choices in preprocessing, model specification, and inference that vary systematically across LLM and persona conditions. Critically, the outcomes are steerable: reassigning the analyst persona or LLM shifts the distribution of results even among methodologically sound runs. These results highlight a central challenge for AI-automated empirical science: when defensible analyses are cheap to generate, evidence becomes abundant and vulnerable to selective reporting. The same capability also helps address it: treating analyst results as distributions makes analytic uncertainty visible, and deploying AI analysts against a published specification can reveal how much disagreement stems from underspecified design choices. Taken together, our results motivate a transparency norm: AI-generated analyses should be accompanied by multiverse-style reporting and full disclosure of the prompts used, on par with code and data.

Read article →

Crowding controls the scaling of bus frequency with demand

2026-07-13 · DOI 10.1073/PNAS.2535998123

Siddharth Patwardhan; Sirag Erkol; Filippo Radicchi; Marc Barthelemy

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Sociales (CNRS/Ecole des Hautes Etudes en Sciences Sociales); Complexity Science Hub; Institut de Physique Theorique, Universite Paris-Saclay, Commissariat a l'energie atomique et aux energies alternatives, CNRS

Abstract. Cities must allocate limited resources to maintain mobility, with uncertainties about the resulting state of the system. Analyzing roughly 3,000 bus routes with more than 4 billion yearly riders across 19 metropolitan areas worldwide, we uncover a robust scaling law of the form f (d / t) with exponent [1 / 2 , 2 / 3] , linking the service frequency f to passenger demand d and route duration t . We show that this scaling emerges from a simple optimization principle: Cities implicitly minimize total passenger waiting time under a fixed operational budget when both schedule frequency and crowding are taken into account. This mechanism produces two universal regimes: a frequency-dominated regime with = 1 / 2 when crowding is negligible and a capacity-dominated regime with = 2 / 3 when most routes are overloaded. Intermediate exponents arise when only part of the network operates near capacity. Furthermore, we find that the benefits of additional investment are highly uneven across systems. For instance, our model suggests that a 20 % budget increase yields nearly a 5-min reduction in daily waiting time per passenger in Boston, compared to only about 1 min in Paris. These findings place urban transit within a broader class of constrained capacity-allocation problems, while highlighting a distinct regime in which prescribed route demands shape the allocation of limited service resources. The resulting scaling laws show how simple optimization principles can generate systematic exponents in complex transport systems, beyond the dissipation-based frameworks usually considered in physical and biological flow networks.

Read article →

Combined signals of scientific and social consensus best mobilize action on societal challenges even if government and industry oppose

2026-07-13 · DOI 10.1073/PNAS.2533408123

Anandita Chakraborty Sabherwal; Gregg Sparkman

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Abstract. When facing real-world societal challenges, people make decisions about how to act based on multiple, often conflicting messages from various influential stakeholders such as governments, scientists, fellow citizens, and private companies. How does the public decide to act amid potentially converging or diverging signals from multiple entities in society? Which entities remain influential even if other groups appear apathetic or stand in opposition? This question becomes especially important in contexts like the United States where the government, industry, scientists, and public sentiment may diverge. In two conjoint experiments spanning over 40,000 evaluations, one behavioral experiment and a digital field study (N s = 1,515, 2,005, 1,604, 50,668 respectively; all preregistered), we show the relative and combined influence of government bodies, scientific experts, industry stakeholders, and social consensus in the general public in driving responses to 10 issues in environmental, health, and technology domains. Across domains, support from scientists and social consensus was most influential in shaping the American public's responses to societal problems including policy support, product preferences, donations to a cause, and social media engagement. And when facing opposition from government and industry, the combined, but not isolated, support of scientists and social consensus reliably mobilized action across domains. This work suggests that even amid rising systemic opposition to solving contemporary societal challenges, scientists and the public can come together to mobilize action.

Read article →

Science

Seabed mining requires timely governance

2026-07-16 · DOI 10.1126/SCIENCE.AEI7769

Carsten Ruhlemann

Federal Institute for Geosciences and Natural Resources (BGR), Hannover, Germany.

Abstract. Abstract unavailable.

Read article →

Science Advances

When trust building looks like surveillance: Public polarization toward Chicago policing

2026-07-17 · DOI 10.1126/SCIADV.AEB7575

Tony Cheng; Shelley Liu

Department of Sociology, Duke University, Durham, NC, USA.; Sanford School of Public Policy, Duke University, Durham, NC, USA.

Abstract. Public debate about policing highlights competing views about whether community initiatives build trust or reinforce surveillance. We argue that perceptions depend on prior state orientations, which can polarize opinions across groups. Using a preregistered survey experiment in Chicago, we examined responses to the police department's "Sentiment Dashboard," a project designed to enhance legitimacy by collecting community feedback. Results show that the dashboard increased trust in police among non-Black respondents but reduced trust among Black respondents, especially when framed as monitoring. Among non-Black respondents, dashboard exposure also reduced support for the Black Lives Matter movement and racial equity policies unrelated to policing, such as diverse hiring and climate justice. Mechanism testing with qualitative evidence suggests that treated non-Black participants were more likely to regard police reforms as substitutes for racial equity policies, rendering further measures unnecessary. These findings reveal how state legitimacy initiatives may instead widen racial divides and weaken coalitions for structural reform.

Read article →

Previous Week Catch-Up

Communication Methods and Measures

Beyond apps: a method to model full distributions of screen content over time to study media effects

2026-07-10 · DOI 10.1080/19312458.2026.2697199

Yikun Chi; Mu-Jung Cho; Kimberly M. Molaib; Xiaoran Sun; Thomas N. Robinson; Nilam Ram; Byron Reeves

Academia Sinica; Stanford University; University of Minnesota-Twin Cities; University of Washington

Abstract. Research on smartphone media use has progressed from aggregate screen-time measures to app-level logs, yet these approaches still obscure the diversity and structure of the content users encounter. We introduce smartphone screen content patterns (content patterns) as a new construct and propose an analytical framework that represents smartphone use as distributions of experienced screen content rather than summaries of time spent on devices or in apps. We operationalize content patterns using deep learning embeddings of moment-by-moment screenshots and formalize use periods as probability distributions over embedding space. This distribution-based representation supports two analytic strategies: Mantel tests to assess covariation between changes in content patterns and outcomes, and k-nearest neighbor regression to evaluate whether content patterns contain sufficient information to reconstruct outcomes. We illustrate the framework with six months of screenomics data from 53 adolescents, including 21 million passively collected screenshots and 542 fortnightly mental health self-reports. Results show that most content variability occurs within applications and time periods, highlighting the limits of app- and time-based measures. Content patterns outperform screen-time measures and provide complementary insights to app-level logs, advancing a content-centric approach to modeling digital media exposure.

Read article →

Introducing HALC: a general pipeline for the systematic and reliable construction of prompts for automated coding with LLMs in the computational social sciences

2026-07-07 · DOI 10.1080/19312458.2026.2693637

Andreas Reich; Claudia Thoms; Tobias Schrimpf

University of Hohenheim

Abstract. LLMs are seeing widespread use for task automation, including automated coding in the social sciences. However, even though researchers have proposed different prompting strategies, their effectiveness varies across LLMs and tasks. Often trial and error practices are still widespread. Our study aims to fill this gap and evaluate how LLMs can be used in a systematic and transparent way to produce reliable codings in content analyses. We propose HALC - a general pipeline that allows for the systematic and reliable construction of prompts for any given coding task and model. We develop this pipeline based on current literature and findings of a prestudy investigating consistency and influencing factors of LLM codings. We also apply HALC on two other datasets covering different thematic contexts, document types, languages, and coding units to test its applicability. Based on more than three million LLM requests, our results demonstrate that the pipeline is capable of identifying prompts for reliable codings in different settings. We also discuss shortcomings and further potential for development.

Read article →

Communication Research

Chatbots and Loneliness: How Linguistic Dynamics in Conversations with Chatbots Shape Young Adults' Loneliness

2026-07-10 · DOI 10.1177/00936502261466012

Jinyoung Choi; Catalina L. Toma

University of Wisconsin-Madison, USA

Abstract. This study theorizes and tests language-based mechanisms through which conversations with chatbots can shape users' loneliness. Lonely college students (N = 96) engaged in emotionally meaningful conversations with the chatbot Replika daily for one week. Results showed a small but significant decrease in users' loneliness from baseline to post-test. Linguistic analyses revealed two distinct well-being pathways. When participants used more positive emotion language, so did the chatbot, which was linked to higher positive daily affect and lower post-study loneliness. Conversely, when participants used more negative emotion language, so did the chatbot, which was linked to lower positive daily affect and higher post-study loneliness. Findings provide evidence for an expression effect , where users' own positive (but not negative) linguistic framing shapes their well-being, and a chatbot-response effect , where chatbots mirror and amplify users' emotional tone. This study theorizes chatbots as emotional mirrors with dual potential to alleviate or exacerbate users' loneliness.

Read article →

Do Profile Photos Enhance Vicarious Intergroup Contact? Testing Theoretical Perspectives from Social Identity, the Social Identity Model of Deindividuation Effects (SIDE), and Social Presence

2026-07-06 · DOI 10.1177/00936502261466759

Yue (Nancy) Dai; Lunrui Fu; Wufan Jia

City University of Hong Kong, Hong Kong; Nanyang Technological University, Singapore; Tsinghua University, Hong Kong

Abstract. Profile photos are ubiquitous in online environments, yet little is known about how they shape observers' impressions when mediated interactions involve members of different social groups. This experiment (N = 1,661) examined how profile photos influence impressions formed during observed intergroup exchanges in news comment sections. A 2 2

design (topic photo) tested mechanisms derived from Social Identity Theory (SIT), the Social Identity Model of Deindividuation Effects (SIDE), and Social Presence Theory. Results showed that profile photos heightened awareness of the relevant group distinction and, through identification with the ingroup commenter, led to more positive outgroup responses-but only when the photos clearly conveyed the group difference central to the interaction. When both interpersonal and intergroup processes were modeled together, only the intergroup pathway remained supported. These findings clarify how profile photos shape mediated impressions and identify cue diagnosticity as key for enhancing vicarious intergroup contact.

Read article →

Journal of Communication

Media emotion regulation and executive functions: individual differences in temporal ordering among young children

2026-07-09 · DOI 10.1093/JOC/JQAG009

Jane Shawcroft; McCall A Booth; Chris L Porter; Megan Van Alfen; Megan Gale; Lara A Maximiano; Elena K Holmes; Sarah M Coyne; Hailey G Holmgren; Blake L Jones

Department of Human Development and Family Studies, Utah State University , Logan, UT,; Department of Psychology, Brigham Young University , Provo, UT,; Department of Psychology, Georgetown University , Washington, D.C.,; School of Communication, The Ohio State University , Columbus, OH,; School of Family Life, Brigham Young University , Provo, UT,

Abstract. Abstract Using media to help children regulate difficult emotions (also called regulatory media use) is a process linked to the development of children's executive functions. Although a necessary condition for establishing causality, the temporal ordering of the relationship between regulatory media use and executive functions has not been explicitly studied. In this study, we examined individual differences in the temporal ordering of the association between regulatory media use and executive functions using data from six waves of Project M.E.D.I.A. (spanning 2.5 years old to 7.5 years old). Three classes of temporal ordering were identified: Media Emotion Regulation Driven (7.00%), Bidirectional (87.36%), and Diverging Temporal Ordering (6.00%). Parent depression was negatively associated with the probability of children being classified in the Bidirectional class but was positively associated with the probability of children being classified in the Media Emotion Regulation Driven class.

Read article →

Nature Communications

The intersecting effects of race/ethnicity, wealth, education, and gender on AIDS among 28.3 million Brazilians

2026-07-09 · DOI 10.1038/S41467-026-73892-6

Iracema Lua; Laio Magno; Andrea F. Silva; Priscila FPS Gestal; Joao Luiz Bastos; Gabriela Jesus; Ronaldo de Almeida Coelho; Maria Yury Ichihara; Mauricio L. Barreto; Carlos AS Teles Santos; Corrina Moucheraud; Pamina M. Gorbach; James Macinko; Luis Eugenio Souza; Davide Rasella; Ines Dourado

Abstract. Abstract unavailable.

Read article →

Quantifying institutional gender inequality in contemporary visual art

2026-07-09 · DOI 10.1038/S41467-026-71470-4

Xindi Wang; Alexander J. Gates; Magnus Resch; Albert-Laszlo Barabasi

Abstract. Abstract unavailable.

Read article →

Urbanization processes widen global divergence in extreme temperature variability

2026-07-08 · DOI 10.1038/S41467-026-75457-Z

Renlu Qiao; Fangzheng Li; Tao Wu; Zeyin Chen; Xiang AO; Xi Meng; Jingkai Zhao; Zhiqiang Wu; Borong Lin; Yue Zhang

Abstract. Abstract Extreme temperature variability (ETV) is a key dimension of climate risk for billions of residents. However, how urbanization shapes global ETV divergence remains unclear. Here, we assemble a 1950-2020 panel of 10,522 cities and demonstrate that ETV trajectories diverge by development status as measured by the human development index (HDI). ETV intensifies in high-development cities, whereas it slowly weakens in low-development cities, resulting in a current gap of roughly 1.66 C. Decomposing ETV into event frequency and intensity reveals that cumulative ETV is driven mainly by intensity. Using an interpretable machine-learning framework, we find that aerosols are the urban factor most strongly associated with ETV after controlling for climate and geography. Blue-green space is consistently associated with lower ETV, whereas urban morphology has a smaller and context-dependent effect. These findings link global ETV inequality to urban governance and support targeted management that focusses on limiting volatility under climate risk.

Read article →

Assessing corporate sustainability with large language models: evidence from Europe

2026-07-07 · DOI 10.1038/S41467-026-75160-Z

Kerstin Forster; Lucas Keil; Victor Wagner; Maximilian A. Muller; Thorsten Sellhorn; Stefan Feuerriegel

Abstract. Abstract Companies play a crucial role in achieving global sustainability goals, yet evidence on their progress across environmental, social, and governance (ESG) dimensions remains limited. We develop a machine learning framework to systematically extract ESG indicators from corporate reports. Applying this approach to annual and sustainability reports of 600 large European firms (2014-2023), we construct a dataset of 2.9 million ESG observations across environmental, social, and governance topics. We assess ESG transparency based on disclosures aligned with the European Sustainability Reporting Standards (ESRS) and evaluate ESG performance using extracted numerical indicators. Results reveal a pronounced transparency gap: firms in the top ESG rating decile disclose 22% more indicators than those in the bottom decile, although this gap narrows over time. Performance trends are uneven: while most social indicators remain largely stagnant, except for gains in gender equality, environmental indicators show some improvement. Reported scope 3 emissions increase sharply, largely reflecting improved disclosure. Our open-source framework enables systematic tracking of corporate ESG efforts.

Read article →

Nature Human Behaviour

Stable intuition and the rise of deliberative prosociality in childhood

2026-07-06 · DOI 10.1038/S41562-026-02487-4

Francesco Margoni; Francesco Nava; Chiara Sotis; Matthew R. Levy; Valerio Capraro; Elena Nava

Abstract. Abstract We examine how social behaviours emerge and stabilize across childhood. A sample of 537 Italian-speaking children (3-10 years) were randomly assigned to respond under time pressure (intuitive) or time delay (deliberative) in social decision tasks (Public Goods, Dictator, Ultimatum, Deception and Moral Dilemmas). Factor analysis identified three latent dimensions: Prosociality (cooperative, altruistic and honest actions), Social Optimism (beliefs about others' cooperation) and Acquiescence (tendency to accept offers). Intuitive responses were more prosocial than deliberative ones in early childhood (= 0.66; 95% CI = (0.35, 0.97)), but this difference diminished with age (= 0.11; 95% CI = (0.18, 0.05)). We found no evidence that Social Optimism varied across age or decision mode, whereas Acquiescence declined with age (= 0.14; 95% CI = (0.19, 0.10)). These findings suggest a developmental shift whereby prosocial behaviour is initially

driven by intuitive responses and gradually becomes embedded within reflective, deliberative decision-making systems, as cooperative dispositions stabilize across childhood.

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Political Analysis

Monotone Ecological Inference

2026-07-10 · DOI 10.1017/PAN.2026.10048

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Abstract. Abstract We study the identification of individual-level associations when only aggregate data are available. We characterize the biases of, and relationships among, canonical ecological inference (EI) estimators. We use these results to develop a partial identification approach: monotone EI. The approach exploits information about one or both of the following conditional associations: (1) outcome differences between groups within the same neighborhood and (2) outcome differences within the same group between neighborhoods with different group compositions. We show how assumptions about the sign of these conditional associations, whether individually or in relation to one another, can yield informative sharp bounds. We illustrate our results using county-level data to study differences in COVID-19 vaccination rates among Republicans and Democrats in the United States.

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Proceedings of the National Academy of Sciences

Multiplexing in networks and diffusion

2026-07-10 · DOI 10.1073/PNAS.2534923123

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Abstract. We study multiplexing-the extent to which people have multiple, distinct types of relationships with the same others. We document empirical multiplexing patterns in Indian village data. We show that relationships such as socializing, advising, helping, and lending are correlated but distinct, while commonly used proxies for networks based on ethnicity and geography are nearly uncorrelated with actual relationships. We also show that these layers and their overlap affect information diffusion in a field experiment. On the theoretical side, we show that increased overlap in people's different types of relationships impedes the spread of simple contagions-such as diseases or basic information that spread through a single interaction. In contrast, such higher levels of multiplexing enhance the spread of a complex contagion (e.g., a behavior or belief that requires multiple interactions for transmission) when infection rates are low, but impede complex contagion if infection rates are high. Finally, we identify empirical differences in multiplexing by gender and connectedness.

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Why human societies adopt rigid moral rules: The efficiency-robustness trade-off

2026-07-08 · DOI 10.1073/PNAS.2535467123

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Abstract. Humans are capable of remarkably flexible moral judgment. Yet societies rely on rigid rules-obligations and prohibitions that apply categorically, even when case-by-case reasoning could yield better outcomes. Why would a species capable of flexibility bind itself to rigid rules? We propose that rigid rules arise as social technologies for managing ambiguity about noncooperation. People often have legitimate reasons for failing to cooperate, yet those reasons are typically opaque to observers, allowing opportunists to disguise selfishness as justified hardship. We formalize this idea with a game-theoretic model. Two cooperative equilibria emerge: a flexible norm that accommodates legitimate excuses but is vulnerable to exploitation, and a rigid norm that closes this loophole by mandating cooperation even when inefficient. Comparing these equilibria reveals an efficiency-robustness trade-off: Flexibility maximizes welfare when trust is secure, whereas rigidity preserves cooperation when trust is fragile. This explains why rigid rules prevail in interactions with strangers, formal institutions, or in tight societies, while flexibility is more common when trust is secure.

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User interaction with digital platforms: A consumer protection perspective

2026-07-06 · DOI 10.1073/PNAS.2525996123

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Federal Trade Commission

Abstract. The rapid growth of digital platforms has transformed how consumers interact with the marketplace and led to new challenges for consumer protection. This Perspective describes how the Federal Trade Commission's Bureau of Economics uses behavioral insights and empirical methods to support efforts to prevent unfair or deceptive business practices in digital markets. We highlight two studies that address critical knowledge gaps in digital consumer protection, including how consumers perceive and respond to advertising disclosures and how digital design choices affect the accessibility and effectiveness of consumer reporting tools. The studies illustrate how behavioral research can be used both prospectively, to evaluate potential remedies, and retrospectively, to assess the effects of past Federal Trade Commission (FTC) actions. The first study is a laboratory experiment that examines how the salience of advertising disclosures influences consumer recognition of and engagement with advertising content, where engagement is defined as the time users spend viewing ads. The second is a retrospective analysis of a redesigned consumer complaint website, using a natural experiment to evaluate the effect of usability improvements on complaint submission rates and information quality. Together, these studies demonstrate how applied research on user behavior can inform marketplace oversight and the design of consumer-facing tools. We conclude by outlining ongoing efforts to expand the role of empirical research at the FTC to strengthen the foundation for evidence-based policymaking in an increasingly complex digital environment.

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Turning policy implications into policy impact: Lessons from behavioral science in financial markets

2026-07-06 · DOI 10.1073/PNAS.2525990123

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Abstract. We describe how research can inform policymaking. We begin by summarizing the policy formation process, offering a framework illustrating how interactions between scientists and policymakers can provide mutual benefits. We then describe four research initiatives from the Securities and Exchange Commission's Office of Investor Research (OIR) as examples of how behavioral sciences can help inform policymakers. The first initiative established a probability-based nationally representative survey panel focused on US investor behavior. The remaining three examined investor communications for mutual fund benchmarks, emerging annuities (known as registered index-linked annuities), and mutual fund fee visuals. We conclude by synthesizing our work and describing practices that can help scientists have policy impact,

not just policy implications.

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The role of nonfinancial factors in the Congressional Budget Office's health insurance coverage projections

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Abstract. Nonfinancial factors play an important role in people's coverage decisions as premiums, cost sharing, and other factors that determine the financial value of health insurance cannot fully explain patterns in take-up rates. The Congressional Budget Office (CBO) is tasked with estimating the budgetary and coverage effects of policy and economic changes that affect health insurance markets, which includes considering the interactions between nonfinancial factors and policy design. The agency groups the nonfinancial factors it considers into three categories: ease of enrollment, awareness, and attitudes. Many recent policies have affected nonfinancial factors related to the take-up of Medicaid and marketplace insurance, such as the introduction of work requirements for Medicaid and access to zero-premium marketplace plans. This article explains how nonfinancial factors are handled and incorporated into CBO's baseline health insurance projections and cost estimates, highlights the recent literature on the effect of these nonfinancial factors on insurance take-up decisions, and suggests future areas of research.

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Science in service: Behavioral research applications across federal agencies

2026-07-06 · DOI 10.1073/PNAS.2608196123

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Abstract. Abstract unavailable.

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Payment integrity in government programs: Takeaways from incorporating the behavioral sciences in US federal evaluations

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Abstract. A primary way the US federal government delivers public goods and services is via monetary payments. Ensuring that these payments are calculated accurately, delivered on time, and made to the correct recipients is important for government fiscal health. Inaccurate or delayed payments can weaken public trust in the government and undermine government accountability. In this article, we examine findings from a set of impact evaluations assessing interventions designed to improve payment integrity in US federal programs. The low-cost, evidence-based interventions draw on insights from the social and behavioral sciences and include modification of forms, changes to how and when agencies request information, and altering existing communications. The evaluations were conducted by the US General Services Administration's Office of Evaluation Sciences in collaboration with agency partners. We extract three takeaways across four

representative evaluations. First, the real-world evaluations validate a key implication of the behavioral science literature: interventions that reduce burdens for individuals have small effects that meaningfully improve payment integrity at scale. Second, effects attenuate across interventions and over time, suggesting a need for iterative evaluation. Finally, bureaucratic hurdles and administrative complexity are the main barriers to translating academic insights into real-world government programs. Addressing these challenges will require close collaboration between behavioral scientists and practitioners throughout the intervention design and evaluation process.

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Feeding health inequality through platform-based food delivery in China

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Abstract. This study examines how the expansion of online food delivery platforms affects health inequality in China. Exploiting the staggered rollout of platforms across counties and nationally representative panel data from 2010 to 2022, we find that platform entry increases overweight among low-income individuals while reducing it among high-income individuals. This divergence operates through two channels: low-income users increase consumption of fried foods and reallocate time from cooking to sedentary leisure, whereas high-income users reduce intake of calorie-dense foods and increase physical activity. These behavioral changes accumulate into downstream health consequences affecting weight-related chronic diseases and children's overweight in low-income households. Equal access to digital convenience thus does not translate into equal health benefits, suggesting that technological diffusion may amplify health disparities.

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Applied behavioral and decision sciences in support of US FDA's drug regulatory mission

2026-07-06 · DOI 10.1073/PNAS.2525995123

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Abstract. The US Food and Drug Administration (FDA) makes and communicates decisions that directly affect the health choices and well-being of the US public. Behavioral and decision scientists have long supported FDA's pharmaceutical (medical drug) regulatory and public health mission, working alongside clinical and pharmaceutical scientists. This Perspective describes four interrelated cases illustrating these collaborations: a) creating the Benefit-Risk Framework to guide regulatory decisions regarding new drug approvals; b) launching an internal decision support service to facilitate specific regulatory decisions; c) implementing the Patient-Focused Drug Development initiative to strengthen the use of patient input to inform regulatory decision making; and d) developing FDA SOURCE, a dynamic systems simulation model to assess potential strategies to address the US opioid overdose crisis. Key to the success of these efforts has been their behavioral and decision science foundations, strategically implemented by an expert team dedicated to fostering trusted collaborations with internal and external partners. These cases offer a model for other agencies facing complex decisions of public impact.

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Science

Why cities cannot learn

2026-07-09 · DOI 10.1126/SCIENCE.AEH8677

Zaheer Allam

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Abstract. For most of the world's urban history, changes in policies and responses to hazards have proceeded on a timescale that made adaptation possible, however slow or politically contested. The Great Fire of London in 1666 produced building regulations within 2 years of the devastating blaze. Repeated flooding events along the Rhine and Danube Rivers through the late 20th century spurred zoning rules and hazard mapping that are now standard across Europe's urban planning systems. What drove these improvements was a general pathway in which disruption produced evidence, evidence generated pressure, and pressure forced modifications by agencies responsible for governing a city's well-being. However, that process has been degrading across every inhabited continent. What has emerged in its place is a volatile urbanism that is threatening resilience.

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A statistical test for the benefits of personalizing interventions

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Abstract. From medicine to marketing to social sciences, the promise of tailoring interventions to individuals is undeniable. However, practical applications force weighing personalization's potential benefits with its possible increased cost and fragility. We introduce a statistical hypothesis test that evaluates, given historical data, evidence that a personalized intervention policy's performance will surpass deploying the best single intervention. The test maintains strict Type I error control while achieving asymptotic normality with the minimal possible variance under specified conditions. Results on diverse datasets from job training, depression treatment, education, and recommendation systems demonstrate the test's versatility and its superior performance over alternatives. This test can support decision-makers throughout the intervention sciences by providing a simple and powerful quantification of the potential benefits of personalization.

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Science Advances

Communicating coral: Art and science for global reef action

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Abstract. Coral reefs are experiencing unprecedented rapid decline. Art can act as a beacon to engage audiences, influence policy, and inspire global action.

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